Measurement Systems

An Open-Source Summary

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April 25, 2026

Contents

> **Note**
>
> This summary will only be based on the slides and lectures. No figures or copy from the cursus book is permitted due to copyrights.
>
> Nonetheless, it is recommended to also go through the cursus book as it may go more in depth than this summary

Part I.

Introduction

1. Introduction

1.1. Why is measuring important ?

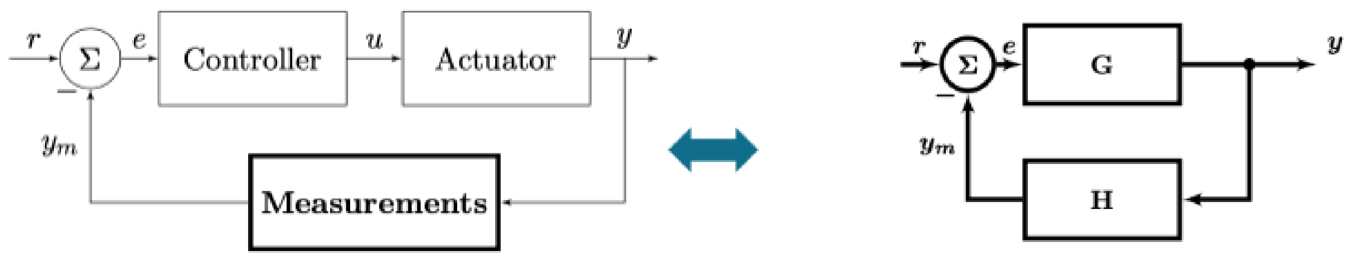

Figure 1: Typical system

In a feedback system, we want some input to transform in a specific output. Later, we want to measure this output to adapt the control and see how effective this is.

Using the standard G and H notation we can find that:

$$y = \frac{G}{1+GH}r \qquad e = \frac{1}{1+GH}r \tag{1}$$

$$y = \lim_{G\to\infty} \frac{G}{1+GH}r \approx \frac{1}{H}r \tag{2}$$

1.2. General principles

> Building a measurement system is transforming energy/information from one domain to another.

We first start by building the physical knowledge (build physical quantity model), mathematical reasoning (find how accurate we can make the model to reflect the system) and finally we must take noise and other limits into consideration.

In a system, at a high-level, there is 2 forms of variables:

1. **Across Variables:** describe the *state*, in **parallel** with the terminal. Effort to change the state
2. **Through Variables:** describe the *flow*, in **series** with the terminal.

Then combining those two can give us some fundamental quantities: - $A \cdot T$: **power** generated or dissipated in the element - A/T: **impedance** of the system. Describes how an element transforms an A into a T

1.3. Modelling a system : the physical model

We usually want to map the IO behavior based on physical knowledge. Each model tries to represent reality but will always introduce errors or not take into account higher order phenomena.

For example, we can model a pressure change using a diaphragm to a capacitance change. If we take a simple spring model we don't take second order term into account which introduces errors.

Table 1: Strength and weaknesses of physical approach

Strength	Weakness
Good insight	Error due to approximations
System optimization is possible	Derivation of this formula is not always straightforward/possible
Calibration techniques can be easily derived	Errors due to tolerances in fabrication

1.4. Correlation

A typical operation used in system is **correlation**, it is used to find the "likelihood" between two signals. It helps us extract features:

$$R_{fg}(t) = \int_{-\infty}^{\infty} f(t)g(t-\tau)d\tau$$

Knowing Fourrier's theory, we can represent any signal as a combination of sin and cos. Instead of using $g(t-\tau)$, this gives us:

$$F(f) = F_{cosine}(f) + F_{sine}(f) \quad (3)$$
$$= \int_{-\infty}^{\infty} (cos(2\pi f t) + isin(2\phi f t))f(t)dt \quad (4)$$

1.4.1. Freshen up of Laplace transform

The Fourier transform can be written as:

$$\mathcal{F}f(t) = \int_{-\infty}^{\infty} e^{i2\pi ft} f(t)dt$$

The Laplace transform goes one step further by using the Euler notation $e^{ix} = cos(x) + isin(x)$ and extend the Fourier transform with complex numbers:

$$\mathcal{L}f(s) = \int_{0}^{\infty} e^{-st} f(t)dt \qquad s = \sigma + i\omega t$$

It can be seen that:

$$\mathcal{F}(f(t)) = \mathcal{L}(f(t)) \Longleftrightarrow f(t) = 0 \text{ if } t < 0$$

Laplace transform is an easy tool to solve differential equation and work with derivative:

$$\mathcal{L}\left(\frac{df(t)}{dt}\right) = s\mathcal{L}(f(t)) + f(0)$$

1.5. The Measurement Chain

> A measurement system converts a state variable in a measured value

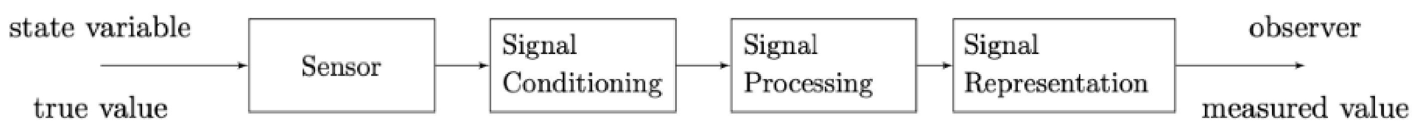

Figure 2: 4 major components

The idea behind a *signal conditioning* is to reduce error and filter out the signal. This should also convert it into a digital signal that can be easily processed by a DSP block. DSP block is used to work with the input and realize complex function while the signal representation can be a screen, plot, …

1.5.1. 4 major components

1.5.1.1. Sensor Each sensor is made of 2 sensors to be exact.

1. A primary that transform a physical quantity to an "*easier to measure*" quantity
2. A secondary that converts this quantity into an electrical signal

1.5.1.2. Signal conditioning This is usually an ADC block. Such block are limited by the current technologies and we call that a **technological constant** C_T. There is no free lunch and everything is a tradeoff leading to a basic formula from information theory:

$$\frac{Speed(Accuracy)^2}{Power} = C_T$$

An ADC needs an input **and** a reference voltage. I outputs $\frac{V_{in}}{V_{ref}} = \frac{N}{2^{n-1}}$. The flash converter using a resistive ladder is a typical (but bad way) to convert into digital. We also need a clock and comparators to convert.

Accuracy is important and the Effective Number of Bits or ENOB is a crucial metric for any system. Due to quantization error and the SNR required for 1 valid bit, we can find such formula[1] :

$$ENOB = \frac{SNDR - 1.76dB}{6.02dB}$$

So in summary, building a physical models is a good step in the right direction to be able to better filter and analyze measurements.

2. Characterization of measurements

2.1. Measurement Errors

In any system, there will be error which will reduce its accuracy. The error is added after each operation in our model but it is important to understand how it **propagates** and its **origin**.

Table 2: Comparing the two type of errors

Characteristics	Absolute error	Relative error
Math	$e = x_m - x$	$e_r = \frac{e}{x} = \frac{x_m - x}{x}$
Unit	Same unit as variable	Unitless - ppm
Estimation	Over-estimation, $e \propto G$	Under-estimation, $e_r ! \propto G$

$$x_m = S(s)G(s)x + G(s)\bar{e_s} + \bar{e_c} \quad \text{Output measurement} \quad \bar{e_m} = G(s)\bar{e_s} + \bar{e_c} \quad \text{Error of this measurement} \tag{5}$$

$$y = \frac{x_m}{S(0)G(0)} \quad \text{Transfer function} \quad \bar{e_i}n = \frac{\bar{e_s}}{S(s)} + \frac{e_c}{S(s)G(s)} \quad \text{Input referred noise} \tag{6}$$

[1] Find more information about how we can get such formula in the DAMSIC course, ***link to a summary of the course***. Check the section 4.1.2

Table 3: Type and difference of errors

Type	Effect	Example
Interfering	Additive to the signal	Magnetic interferences
Modifying	Modifies transfer function	Pressure sensor changes over deformation

By understanding the nature of an error, we can reduce it or even make it disappear. Example, EM interference can disrupt ESG signals. Solution, add an extra probe (right leg) to also capture this interference and connect it to the V_{ref}. Now, V_{in} and V_{ref} in the ADC will have the same error which will cancel out.

2.2. Error Propagation

We must understand how an error propagates in a system. We can build a simple model like:

$$x_m = ax + by$$

Where x is the true value that will be measured x_m, but also possible interferance such as y. The error is:

$$e_m = a\Delta x + b\Delta y$$

We can also conduct some statistical analysis:

$$Var(x_m) = \mathbb{E}\left[(ax+by) - (\mathbb{E}(ax+by))^2\right] \tag{7}$$
$$= \ldots \tag{8}$$
$$\sigma_m^2 = a^2\sigma_x^2 + b^2\sigma_y^2 + 2ab\sigma_{xy} \tag{9}$$

The last term is often forgotten because a lot of person assume $\perp$ between x and y which is not always the case. The covariance is null only if uncorrelated. So typically, if the errors are due to the same source, their covariance is most likely correlated.

2.2.1. Non-linear functions

For more advance transfer function that are non-linear, we can rewrite it using the Taylor series:

$$f(x,y) = f(x_1,y_1) + \frac{\partial f}{\partial x}(x-x_1) + \frac{\partial f}{\partial y}(y-y_1) + \ldots$$

Assuming a small error we can simplify our function and say that

$$\sigma_m^2 \approx \frac{\partial f}{\partial x}^2\sigma_x^2 + \frac{\partial f}{\partial y}^2\sigma_y^2 + 2\frac{\partial f}{\partial x}\frac{\partial f}{\partial y}\sigma_{xy}$$

2.2.2. Static Characteristics

- **Range & Span**: range is the min and max while the span is the delta between the two
  - Those values should reflect where the system is still meeting its specifications

2.2.2.1. Linearizing the reading

Knowing the range, we could force a simple linearization using:

$$O - O_{min} = \frac{O_{max} - O_{min}}{I_{max} - I_{min}}(I - I_{min})$$

But this is far from ideal, a better solution is the least square solution:

$$K = \frac{Cov(X_1, Y_1)}{Var(X_1)} \tag{10}$$

$$= \frac{X_1 Y_1 - n\overline{X_1 Y_1}}{X_1 Y_1 - n\overline{X_1}^2} \tag{11}$$

Table 4: Quick overview of the metrics

Metric	Description	Issue
Max error	Maximum error from ideal and model	Doesn't give a real insight
DNL	Difference between width of the step and ideal LSB step. <1LSB or problematic	Watch out for sign inversion
INL	Maximum deviation (max(DNL))	Less insight
THD	Each non-linearity causes distortion, highlight the distortion nature	Doesn't give information about the non-linearity characteristic

It is important to understand the error of the non-linearity. Because compensating for it is costly as we will use a polynomial to cancel it. The order of the polynomial will require $n + 1$ calibration measurements.

2.2.2.2. Resolution

> **Definition**
>
> Resolution is the smallest discrete step a system can take
>
> $$resolution = \frac{\Delta I_R}{I_{max} - I_{min}} \cdot 100\%$$

The resolution isn't the accuracy as we could add dummy 0 which won't improve the accuracy but the resolution.

2.2.2.3. Hysteresis It is a truly physical phenomena that depicts how when changing the input and reducing it will not lead to the same output path due to various reasons.

2.2.2.4. Dynamical errors When measuring something, we often think we are in the steady state which is not really the case. Most of the time we will wait x amount of time until it is "*good enough*".

This is all linked with the **memory elements** that is present in mechanical, electrical, $\ldots$ systems. The more **independent** memory elements we have the higher is the order of the system. We talk here about settling time which models the *exponential* behavior of such event;

$$\varepsilon = \Delta I expt\left(-\frac{t}{\tau}\right)$$

Of course, Taylor series can be used for small amplitude: $TF(s) = \frac{\partial O}{\partial I}|_{I=I^*} 1/(1+s\tau)$

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2.3. Measurement Characteristics

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2.3.1. Quantization noise

When measuring through an ADC, we can see the loss of precision due to limited bitwidth as noise. It is not physical noise but more like a mathematical tricks to model the impact of quantization on signal. For this to work we must use the postula that the input is a varying signal. If it is not a varying signal, the error will be constant and not looking like white noise aka spread equally. The mean value of this noise will be 0:

$$\mathbb{E}(e_q) = \frac{1}{\Delta I_R} \int_{-\frac{\Delta I_R}{2}}^{\frac{\Delta I_R}{2}} e\, de = 0$$

The power can be seen as:

$$\begin{aligned}
\sigma_q^2 &= \frac{1}{\Delta I_R} \mathbb{E}\left((e_q - \mathbb{E}(e_q))^2\right) && (12) \\
&= \frac{1}{\Delta I_R} \int_{-\frac{\Delta I_R}{2}}^{\frac{\Delta I_R}{2}} e^2\, de && (13) \\
&= \frac{\Delta^2 I_R}{12} && (14)
\end{aligned}$$

As used and explained earlier, the σ^2 is the power of the signal (total integrated noise). The spectrum is limited to $f_s/2$. So we can model an ADC with an extra white noise source with a noise power of $\frac{\Delta^2 I_R}{12}$ with a bw of $f_s/2$.

3. Building a chain : reducing errors

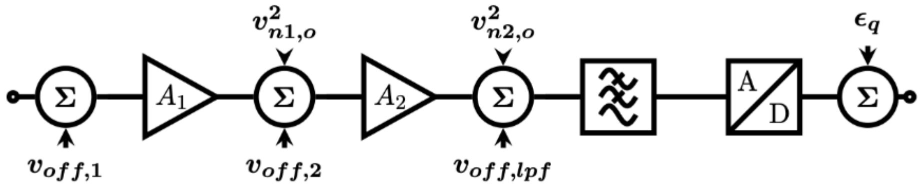

Figure 3: Measurement chain

Each blocks add offset, gain error and noise. This is commonly represented as the chain above.

3.1. Gain distribution

A common way to summarize a chain is by using *input referred* value. This also showcases some basic idea in measurement chains.

$$v_{in}^2 = \frac{v_{n1,o}^2}{A_1^2} + \frac{v_{n2,o}^2}{A_1^2 A_2^2} \qquad I_{ADC} = (V_{in} + v_{off,1})A_1 A_2 + v_{off,2} A_2$$

We have to maximize the first stage gain as it will reduce the impact of noise for later stage components. But we are limited by the offset in the ADC.

3.2. Noise filtering

Another way to reduce the noise by averaging it:

$$\sigma_n^2 = \sum_i^N \frac{x_i - \bar{x}}{N-1}$$

This corresponds to filtering in the frequency domain. We have to be careful with aliasing and this technique cannot refer the $1/f$ noise.

3.3. Chopping and modulation

Offset and $1/f$ cannot be filtered sadly. We would need 0 bandwidth filter for this. Another solution would be to **shift** the noise to higher frequencies:

$$sin(\omega t) sin(\omega_c t) = \frac{cos((\omega_c - \omega)t) + cos((\omega_c + \omega)t)}{2}$$

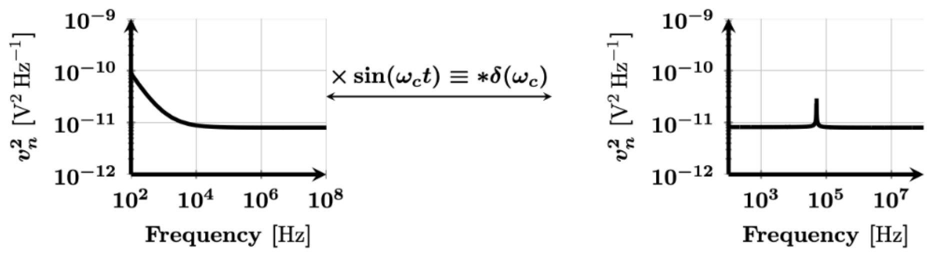

Figure 4: Chopping

3.3.1. Demodulation

Using a sine, half of the energy is lost sadly. This is why actual chopping with a square wave is better.

3.3.2. Chopping

$$S(t)S(t)|_{\omega=0} = \left(\frac{4}{\pi}\right)^2 \sum_{n=0}^{\infty} \frac{1}{(2n+1)^2} \sin\left(2\pi(2n+1)\frac{t}{T}\right) \sin\left(2\pi(2n+1)\frac{t}{T}\right) = \frac{8}{\pi^2}\frac{\pi^2}{8} = 1$$

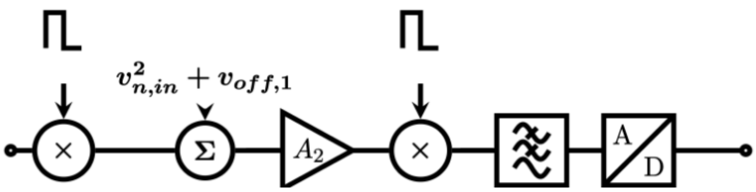

Figure 5: Chopping chain - with representation of the tones (no need to remember the math)

This will retain the signal energy. The input signal is modulated with pulse train, offset + $1/f$ added by A_2. Then demodulated at higher frequencies. Thus, a low-pass filter will not take the high frequency $1/f$ and offset. We are using a 1 -1 square wave.

We must have perfect reconstruction but amplifier has a finite bw. This cannot be easily realized as real components are not perfect. Chopping error.

The square wave signal contains all of the frequency and some beyond the amplifier A_2. This result also in chopping spikes. The chopping error at $2\omega_c$ will be gone but still problems. Energy put at $2\omega_c$ is also lost at the 0 frequency. It is a gain error, if 0 at input, 0 at output no error. The stronger is the signal the larger is the error thus gain error. The error becomes:

$$\frac{4\pi}{\sqrt{2}}\sqrt{\left(\frac{\tau}{T}\right)^2 \frac{1}{1+\frac{\tau}{T}}}$$

3.4. Feedback

A good solution is feedback as it will remove distortion and improve settling.

$$y = \frac{MG}{1+MGH}V_{in} + \frac{G}{1+MGH}\varepsilon \approx \frac{V_{in}}{H} + \frac{\varepsilon}{MH} = \frac{V_{in}}{H} \quad G \to \infty$$

3.4.0.1. Example Measurement of current in electric car. Use a TMR system but has 2 issues:

1. hysteresis
2. temperature issue: modifying environment error

We cannot directly measure on the path, add coils to counter-act the magnetic field which will linearize the reading by fighting an EM field effect with another EM field.

3.5. Calibration

> **Definition**
>
> Calibration is a method to remove an error by measuring the signal with a better sensor and using this measurement to compensate errors (typically drift)

Typically realize offset and gain calibration. This can be realized in the factory, online (using a switch that measure known value then compare. Can also be a random signal in more advanced system) or timed calibration.

If we are measuring and calibrating with a worse system, we are increasing our error.

3.5.0.1. Gain error example

$$V_{out,cal} = (BS + V_{offset} - a)b \qquad \hat{a} = b(a - V_{offset}) \tag{15}$$

$$V_{out,cal} = B\hat{b} + \hat{a} \qquad \hat{b} = Sb \tag{16}$$

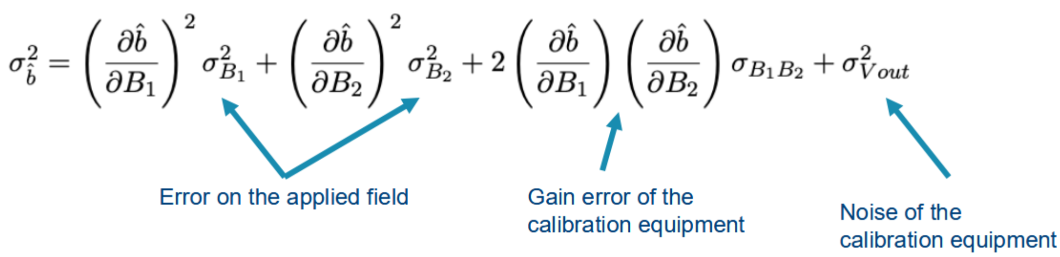

Figure 6: The calibration is as good as the calibration equipment

We need two measurements to be able to model the error. We can see in the math that our calibration will be as good as the accuracy and noise we have.

Need long time to measure to average out the noise. Know the curve and what we should be expecting. Spread the points as wide as possible. A small error between two close points will result in large issues. Calibration is thus a mean to remove *stochastic error*.

3.6. Compensation

> **Definition**
>
> Compensation is a technique where an environmental error is removed by measuring th is variable using another sensor

3.6.1. Hall sensor compensation

Typically, if we know that a sensor may be subjected to variation due to temperature, we will try to measure temperature too to counteract this effect.

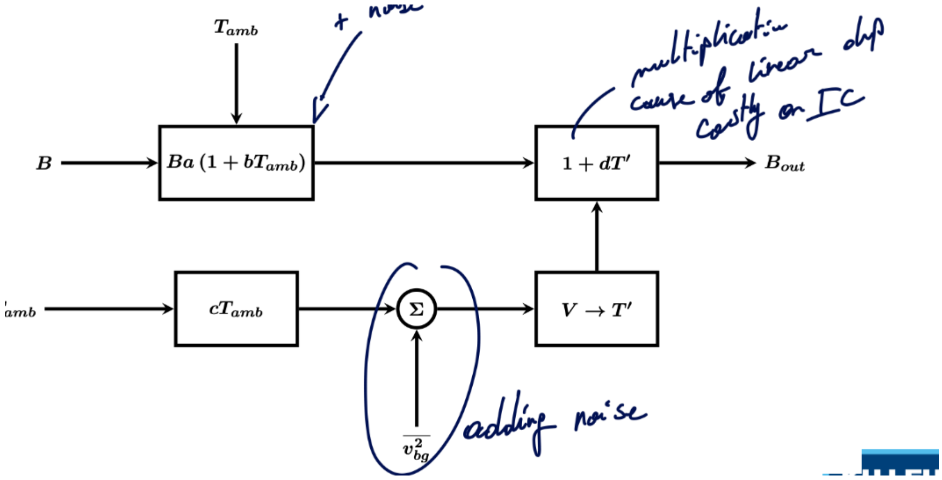

Figure 7: Compensation of a Hall sensor

> *Note*
>
> Here, we multiply T' by d instead of dividing it as division is an expensive operation in digital.

The output of this chain is:

$$B_{out} = Ba(1 + bT_{amb} + cdT_{amb} + bcdT_{amb}^2)$$

We want the B_{out}/B ratio to be constant, in other words not dependent on the temperature. One solution is to make the slope 0 at some temperature. This will require **2** calibration points.

We could also use a quadratic measurement chain. This can make the slope equal 0 on a larger interval of temperature.

3.6.1.1. noise of the temperature sensor But, we will also have noise coming from this compensation chain. Namely the added noise is modeled as:

$$\sigma_{B_{out}}^2 = \left(\frac{dB_{out}}{dT'}\right)^2\Bigg|_{T'=T_{amb}} \sigma_{T'}^2$$

Luckily, the temperature is a slow varying process meaning that the bandwidth is in the Hz range. This can be thus filtered out as magnetic field has a bandwidth in the kHz range.

3.7. Conclusion

- White noise can be filtered
- Flicker noise limits the applicability of filtering
  - Move 1/f to higher frequencies : chopping
- Static errors can be removed using calibration
- Environmental errors can be removed using compensation
- BUT : every correction technique adds errors
  - Filtering $\rightarrow$ settling error
  - Chopping $\rightarrow$ gain error
  - Calibration $\rightarrow$ as good as calibration equipment
  - Compensation $\rightarrow$ add noise of compensating sensor

4. Measuring electrical quantities

4.1. OpAmp

Basic facts about OpAmp:

- High input impedance: so almost no current flows in
- Low output impedance: delivers lots
- Very high gain
- Has a positive and negative terminal

- Non-inverting amplifier

$$v_{fb} = \frac{R_1}{R_1 + R_2} v_{out}$$
$$A(v_{in} - v_{fb}) = v_{out}$$

$$\frac{v_{out}}{v_{in}} = \frac{A\left(R_1 + R_2\right)}{R_1 + R_2 + AR_1}$$

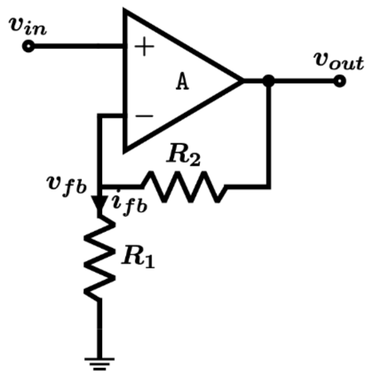

Figure 8: Non-inverting amplifier

With $\lim_{A \to \infty}$ we get $1 + \frac{R_2}{R_1}$. This means we can control the exact gain by controling the resistance ratio. Then, by combining two together, we can make an *instrumentation amplifier*:

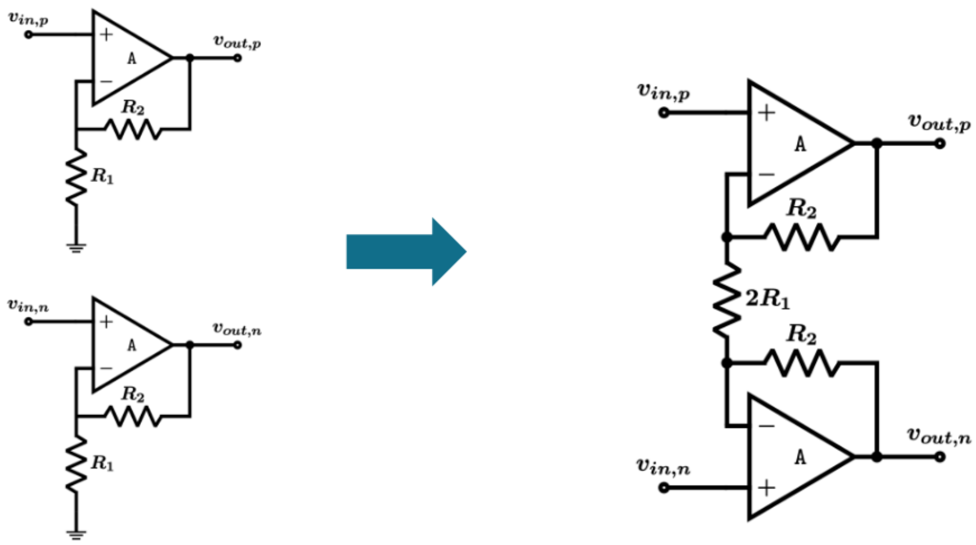

Figure 9: Instrumentation amplifier

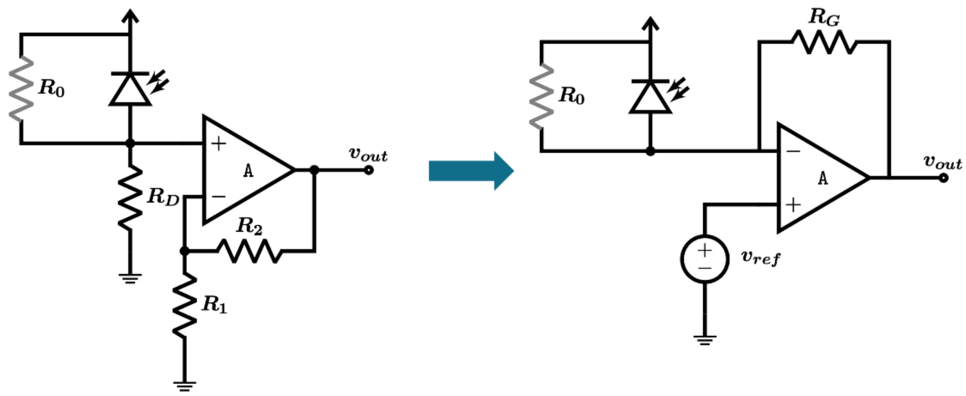

Figure 10: Measuring the current

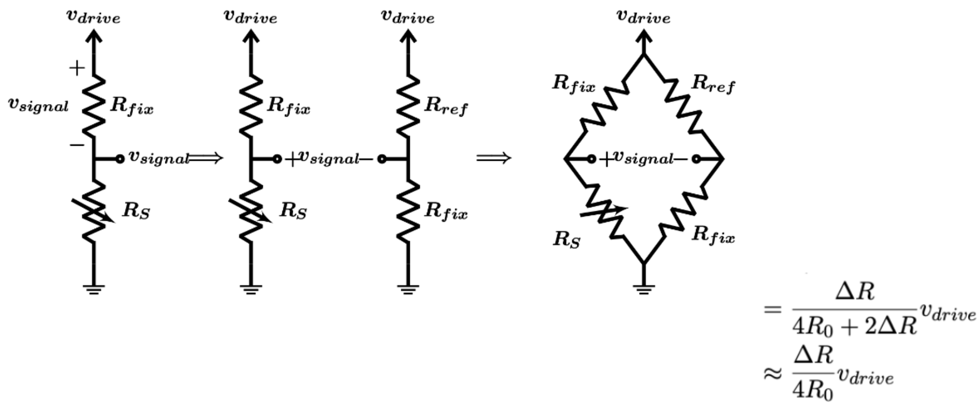

Figure 11: Wheatstone bridge

In the Wheatstone bridge, the $2\Delta R$ factor at the denominator will create non-linearity.

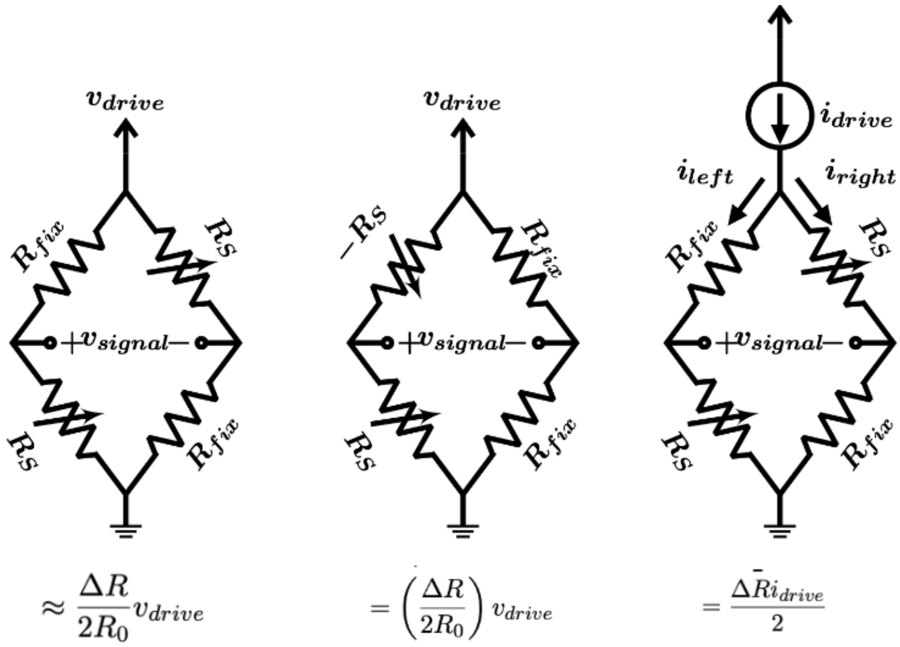

Figure 12: Improvements

4.2. Measuring reactive elements

As seen previously, reactive elements cannot be measured with DC currents as they will either short or be open. A good way is to apply a chopping/square signal instead of a sin to avoid the reduction of energy by a factor 2.

4.2.1. Using oscillators to measure reactive elements

An oscillator is a feedback device that becomes a positive feedback. This must respect the **Barkhausen criterium**:

$$|G(j\omega_n)H(j\omega_n)| = 1 \tag{17}$$

$$\angle G(j\omega_n) + \angle H(j\omega_n) = \pi \tag{18}$$

The π indicates an inverse of the sign. Instead of a negative feedback, it becomes positive.

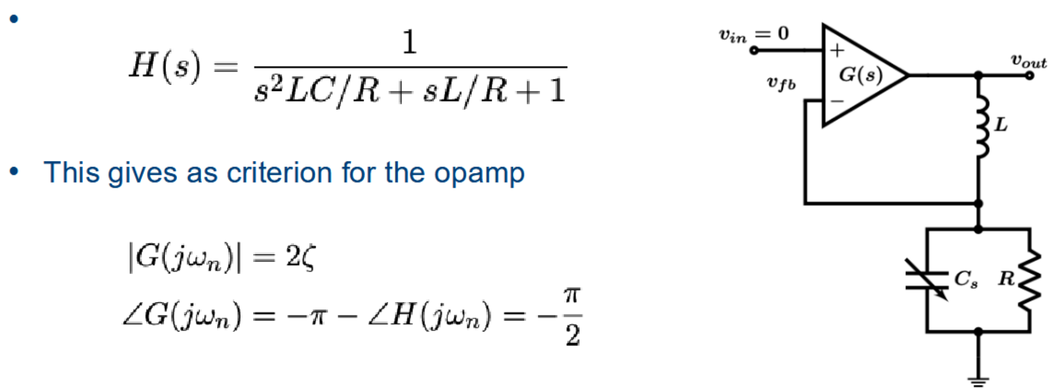

Figure 13: Measuring reactive elements

The resistance must be compensated.

$$\omega_n = \frac{1}{\sqrt{LC}} \qquad \zeta = \frac{1}{2R}\sqrt{\frac{L}{C}} \qquad Q = R\sqrt{\frac{C}{L}}$$

The Q factor is a metric that depicts the quality of a system. It is the ratio of the total energy of the system over the power dissipated over a cycle ω:

$$Q = \frac{1}{2\zeta} = \omega \frac{\text{Max energy stored}}{\text{Power loss}}$$

The higher is the Q factor, the narrower is the oscillation band. This makes it tough to oscillate but once in oscillation, it is hard to stop it.

4.2.1.1. Measuring the frequency We can use a counter based on a system clock that will monitor the oscillation. The precision of this measurements is as a good as the digital clock (jitter, noise, …). Of course, the clock should have a higher frequency than the frequency we are trying to monitor (Shanon sampling frequency, …).

$$\begin{aligned} \Delta\omega &= \left(\frac{\partial \omega_n}{\partial C}\right) \Delta C && (19) \\ &= \frac{1}{2}\frac{1}{\sqrt{LC^3}}\Delta C && (20) \\ &= \frac{\omega_n}{2C_0}\Delta C && (21) \end{aligned}$$

4.3. Conclusion

- The design of a measurement chain is aimed to reduce the errors :
  - Meet the input range of the ADC to reduce quantization noise
    - to avoid clipping or simply not using the full dynamic range
  - Filter of noise
  - Use chopping / modulation to be able to split frequencies
  - Calibration can be used to reduce production errors
    - For stochastic
  - Compensation to reduce environmental errors
  - Use feedback to reduce errors and improve dynamics

Part II.
sensors and specific measurement equipment

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Part III.
Exam Questions

5. Theory questions

NotebookLM but verified by hand:

Question 1: Discuss resolution in measurement systems, give 2 examples. What is the influence on noise?

Answer: Resolution is defined as the smallest change in the input signal that produces a detectable change in the output signal.

- **Example 1:** A wire-wound potentiometer has a resolution limited by the thickness of the wires; the wiper moves in discrete steps from one turn to the next.
- **Example 2:** An Analog-to-Digital Converter (ADC) has a resolution defined by its bit depth. For a voltage range V_{ref}, the resolution is $V_{ref}/2^n$.
- **Influence on noise:** Resolution is often modeled as **quantization noise**. This is a mathematical formulation of the quantization error and is not a real noise! We assume random varying input which will create an uniformly distributed error. This error is assumed to be uniformly distributed between ± 0.5 LSB, resulting in a noise variance (power) of $\Delta I_R^2/12$, where ΔI_R is the resolution step.

Question 2: Describe hysteresis. Give one example. Explain how feedback can be used to reduce hysteresis.

Answer: Hysteresis is a static error where the output value depends on the direction of the input change (history dependence). A common example is magnetic hysteresis in ferromagnetic materials, or mechanical backlash in a gearbox. Feedback reduces hysteresis by placing the non-linear element in the forward path (G) and having a high loop gain. The transfer function becomes $y \approx 1/H$, meaning the system accuracy depends on the feedback path (H) rather than the hysteresis-prone forward path. For example, in a current sensor, a feedback coil generates a magnetic field to cancel the field being measured, keeping the hysteresis-prone magnetic sensor at a constant zero-field operating point.

Question 3: Elaborate on non-linearity. Which measurement characteristics describe non-linearity. Explain how feedback can be used to reduce non-linearity.

Answer: Non-linearity is the deviation of a system's transfer function from a straight line.

- **Characteristics:** It is described by the maximum deviation from an Ideal Straight Line (often using least-square approximation), Integral Non-Linearity (INL), Differential Non-Linearity (DNL), and Total Harmonic Distortion (THD) in the frequency domain.
- **Feedback:** Similar to hysteresis, negative feedback linearizes a system. If the forward gain G is sufficiently high, the closed-loop transfer function approximates $1/H$. If the feedback component H is linear, the overall system becomes linear, regardless of the non-linearity in G.

Question 4: Discuss the difference between rise time and settling time for a second order measurement system.

Answer:

- **Rise Time (t_r):** The time required for the response to rise from a specified low value (e.g., 10%) to a specified high value (e.g., 90%) of its final value, or simply the time to reach the first crossing of the final value.

- **Settling Time (t_{settle}):** The time required for the response curve to reach and stay within a specified error band (e.g., $\pm 2\%$) of the final steady-state value. In an underdamped second-order system, the signal rises quickly (short rise time) but oscillates (rings), potentially causing a long settling time.

Question 5: Discuss the importance of differential systems. Discuss non-linearity and environmental interferers.

Answer: Differential systems measure the difference between two signals ($I_+ - I_-$) rather than a single signal relative to the ground.

- **Non-linearity:** In a differential system, the even-order harmonic terms in the Taylor series expansion cancel out. This eliminates even-order harmonic distortion, leaving only odd harmonics, which improves linearity.
- **Environmental Interferers:** Interfering inputs (like electromagnetic noise or supply voltage ripples) often affect both signal lines equally (Common Mode). A differential amplifier rejects these common-mode signals, amplifying only the differential measurement signal.

Question 6: Explain aliasing. Couple this to kT/C noise. When having a resistive sensor of 1/(2pi) MOhm filtered with a capacitor of 100pF, why would we need to take kT/C when we sample at 1kHz?

Answer: Aliasing occurs when a signal is sampled at a frequency f_s less than twice its bandwidth. Frequencies above $f_s/2$ fold back into the baseband, causing distortion. kT/C noise is the total integrated thermal noise on a capacitor. Even if a low-pass RC filter has a bandwidth much higher than the sampling frequency, the wideband thermal noise above $f_s/2$ will alias (fold) down into the baseband during sampling.

- **Example:** With $R \approx 159k\Omega$ and $C = 100pF$, the filter bandwidth is $f_c = 1/(2\pi RC) \approx 10kHz$. If we sample at 1kHz, the noise between 0.5kHz and 10kHz (and beyond) folds into the 0-500Hz band. The total noise power remains kT/C regardless of the sampling rate, making it the fundamental noise limit for sampled systems.

Question 7: Long term drift. Explain why we can accelerate the drift measurements. Give 2 different drift tests.

Answer: Long-term drift is caused by aging processes (chemical/mechanical) which are temperature-dependent. According to the Arrhenius equation, reaction rates increase exponentially with temperature. Therefore, testing at elevated temperatures accelerates the aging process, allowing lifetime prediction in a shorter time.

- **Tests:**
  1. **HTOL (High Temperature Operating Life):** Device operates at high temp (e.g., 150°C).
  2. **TCT (Temperature Cycling Test):** Device is cycled between temperature extremes to test mechanical stress robustness.

Question 8: Explain the difference between white noise and pink noise. Explain why we cannot filter 1/f noise.

Answer:

- **White Noise:** Has a flat power spectral density (energy is constant per Hz). Originates from thermal motion of electrons.
- **Pink Noise (1/f):** Power spectral density is inversely proportional to frequency (increases at low frequencies). Originates from carrier trapping/detrapping.
- **Filtering:** Standard low-pass or high-pass filters cannot remove $1/f$ noise because it dominates at low frequencies (near DC), which is typically where the measurement signal of interest resides. Filtering the noise would also filter the signal.

What about chopping ???

Question 9: Why can we represent resolution as quantization noise? Can we filter quantization noise? Elaborate.

Answer: Resolution error is the rounding error between the analog input and digitized output. This error is uniformly distributed between ± 0.5 LSB. Statistically, this acts like an added noise source with variance $\sigma_q^2 = \Delta^2/12$.

- **Filtering:** Yes, quantization noise can be filtered. Since it is approximately white noise spread over the Nyquist bandwidth ($f_s/2$), sampling at a higher frequency (oversampling) and then digitally low-pass filtering (averaging) reduces the noise power in the signal bandwidth, effectively increasing the effective number of bits (ENOB).
- Typically we can use some sigma delta design to oversample and spread the noise.

Question 10: Give a block diagram of a generalized measurement chain. Discuss gain distribution. Why would we sometimes split up the gain in several amplifiers in series?

Answer:

- **Block Diagram:** Sensor $\rightarrow$ Signal Conditioning (Amplification/Filtering) $\rightarrow$ Signal Processing (DSP) $\rightarrow$ Representation.
- **Gain Distribution:** To minimize noise, high gain should be applied as early as possible in the chain. However, high gain on the first stage can cause saturation due to the amplification of offset voltages or large DC components.
- **Splitting Gain:** Gain is split into multiple stages (e.g., Stage 1 $\rightarrow$ Filter/Offset Removal $\rightarrow$ Stage 2) to amplify the signal without saturating the chain with offset errors. Typically if each OpAmp has 1 V of output swing, we could quickly saturates if we have a micro V measurement and mV offset.

Question 11: Explain chopping. Why would we use this in measurement systems? What is the origin of chopping ripple?

Answer: Chopping is a modulation technique where the input signal is modulated (multiplied) by a square wave to a higher frequency. The amplifier then processes this high-frequency signal. Since the amplifier's offset and $1/f$ noise are at low frequencies, they are separated from the signal.

- **Use:** It allows the removal of offset and $1/f$ noise by filtering or demodulation later in the chain.
- **Ripple Origin:** Chopping ripple arises because the square wave modulation produces harmonics. If the amplifier has finite bandwidth or if the subsequent low-pass filter is not perfect, residual high-frequency components remain, causing ripple.

Question 12: What is aliasing? How can we avoid this? What is the consequence for a measurement system?

Answer: Aliasing is the phenomenon where high-frequency signals (above the Nyquist frequency $f_s/2$) are indistinguishable from lower frequencies after sampling.

- **Avoidance:** Apply an anti-aliasing low-pass filter before the ADC to attenuate frequencies above $f_s/2$.
- **Consequence:** High-frequency noise or interference folds into the signal band, corrupting the measurement and increasing the noise floor (e.g., kT/C noise).

Question 13: Discuss calibration: what will limit the accuracy? why would we need to take a long calibration time into account? Give an example why knowledge on the sensor helps the calibration process.

Answer:

- **Limit:** The accuracy of a calibrated sensor is limited by the accuracy of the reference equipment used for calibration.

- **Time:** Calibration measurements must be averaged over a long time to filter out noise, ensuring that only the systematic error (offset/gain) is corrected.
- **Knowledge:** Knowing the physics allows using the correct model (e.g., polynomial order). For example, knowing a thermistor follows an exponential law allows for better linearization and fewer calibration points than blindly fitting a high-order polynomial.

Question 14: Explain the difference between compensation and calibration.

Answer:

- **Calibration:** Correcting the device's intrinsic imperfections (tolerances like offset and gain error) by comparing it against a known standard. It is a one-time or periodic factory process.
- **Compensation:** Correcting for *external* environmental effects (like temperature or stress) by measuring that environment (e.g., using a temp sensor) and adjusting the output in real-time based on a known sensitivity model.

Question 15: Draw the schematic of an instrumentation amplifier. Elaborate on its most important properties.

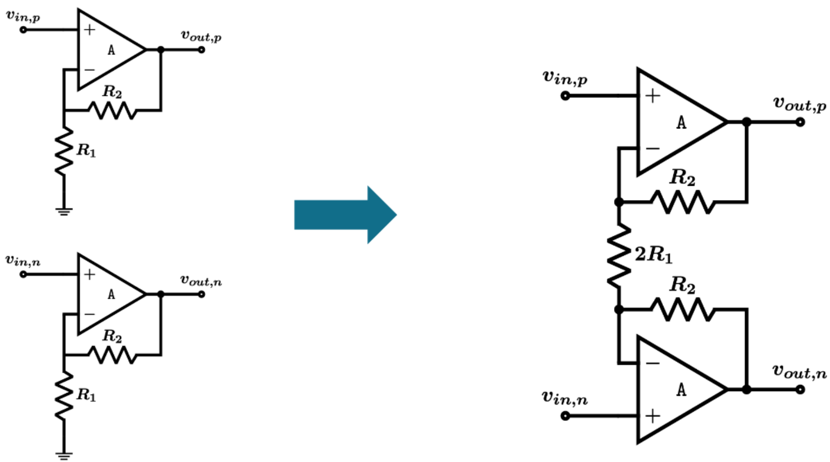


Figure 14: Instrumentation Amplifier

Answer: An instrumentation amplifier consists of two input buffers (non-inverting amplifiers) feeding into a differential amplifier.

- **Properties:** It offers very high input impedance (does not load the sensor), high differential gain (set by a single resistor R_g), and high Common Mode Rejection Ratio (CMRR) to reject environmental noise.

Question 16: Explain the non-linearity of a Wheatstone bridge with only 1 sensing element. Discuss how to improve this. Give an example on how we could create a measurement system with 2 resistive sensors with inverted sensitivities.

Answer:

- **Non-linearity:** A quarter bridge (1 variable resistor) has a transfer function $V_{out} \approx V_{drive} \frac{\Delta R}{4R_0+2\Delta R}$. The term $2\Delta R$ in the denominator causes non-linearity.
- **Improvement:** Use a constant current drive or use multiple sensing elements. Using two elements with opposite sensitivities (one increases R, one decreases R) in the same branch linearizes the output.
- **Example:** A bending beam with one strain gauge on top (tension) and one on the bottom (compression) creates opposite resistance changes.

Question 17: Describe the design of an oscillator as a sensor readout interface for a reactive sensor element.

Answer: A reactive sensor (Capacitor or Inductor) is placed in the feedback network of an amplifier to form an oscillator. The circuit must satisfy the Barkhausen criteria (Loop gain ≥ 1, Phase shift = $360°$). The oscillation frequency becomes a function of the sensor value (e.g., $\omega_n = 1/\sqrt{LC}$). The readout measures the frequency or period, effectively digitizing the signal without a standard ADC. We just need a digital counter.

Question 18: Explain the working principle of a strain gauge.

Answer: A strain gauge is a resistive sensor where resistance changes due to mechanical deformation (strain). The resistance change $\Delta R/R$ is caused by changes in geometry (length l increases, area A decreases) and the change in resistivity ρ. It is characterized by the Gauge Factor G usually between 2 to 2.2.

Moreover, the stress and strain have a linear relationship. It is sensitive in only 1 direction

Question 19: Explain the working principle of an Hall sensor. Bonus question: explain spinning and explain why we can spin a Hall sensor and not another Wheatstone bridge.

Answer: A Hall sensor uses the Lorentz force. When current flows through a conductor in a magnetic field, charge carriers are deflected, creating a voltage perpendicular to both current and field. A hall sensor can be modeled as a wheatstone bridge which is anti-symmetrical. This is a major difference between regular wheatstone bridge which have one sensing arm and one reference arm, here everything is sensing and this fact is used for spinning.

- **Spinning:** A technique to remove offset. The current direction is rotated by 90 degrees (terminals are swapped). The signal polarity flips, but the offset (due to resistive mismatch) remains or behaves differently. Averaging the phases cancels the offset. This is possible because the Hall plate is a symmetric 4-terminal device where input and output ports are geometrically interchangeable, unlike a standard Wheatstone bridge which has distinct drive and sense nodes.

Question 20: Why do we need at least 2 temperature sensors for getting a high accuracy in a MoX sensor?

Answer:

1. **Heater Control:** One sensor measures the heater temperature to maintain the precise high temperature required for the chemical reaction. We must precisely heat up MOx to 600 degree C.
2. **Compensation:** The second sensor measures the temperature of the sensing gas layer itself to compensate for the temperature dependence of the MOx material's resistivity.

Question 21: Explain a capacitive pressure sensor. Why do we need a CDAC?

Answer: A capacitive pressure sensor consists of a fixed plate and a flexible diaphragm. Pressure bends the diaphragm, changing the distance d and thus the capacitance $C = \epsilon A/d$.

- **CDAC:** A Capacitive Digital-to-Analog Converter is needed to subtract the large resting capacitance C_0 of the sensor. The sensor measures $C_0 + \Delta C$. Since $C_0 \gg \Delta C$, amplifying the raw signal would saturate the readout. The CDAC cancels C_0 so only ΔC is amplified.

Question 22: Explain how an accelerometer works. What is the difference between a capacitive accelerometer and a piezoresistive based accelerometer.

Answer: An accelerometer uses a mass-spring-damper system. Acceleration acts on the mass ($F = ma$), causing a displacement x against the spring.

- **Capacitive:** Measures the displacement x by detecting the change in capacitance between fingers on the mass and a fixed frame. It has low power and low noise. Needs a CDAC for accurate readout.
- **Piezoresistive:** Measures the stress in the spring/beam supporting the mass using strain gauges. It is generally less sensitive and consumes more power (resistive heating) but is easier to read out.

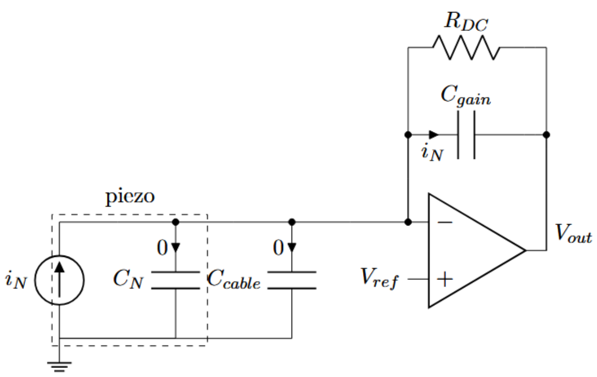
Question 23: Give 3 examples of capacitive sensors: one where we measure a difference in distance, one where we measure a difference in dielectric, one where we measure a difference in area.

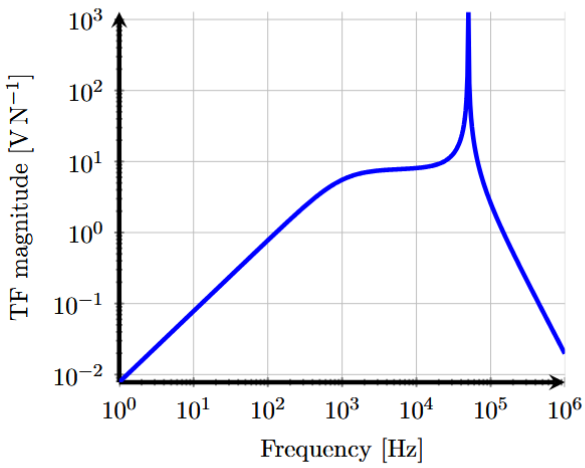
Answer:

1. **Distance:** Pressure sensor (diaphragm moves).
2. **Dielectric:** Humidity sensor (polymer absorbs water, changing ϵ_r).
3. **Area:** Capacitive touch screens.

Question 24: Draw a typical piezo crystal readout. Explain the bode plot of a typical piezo crystal readout.

Answer:





- **Readout:** Typically a charge amplifier (op-amp with a capacitor in the feedback loop) to convert the charge q generated by the crystal into a voltage.
- **Bode Plot:** A piezo sensor behaves like a second-order system. The Bode plot shows a flat response (or rising depending on the variable) up to a sharp **resonance peak** at the crystal's natural frequency ω_n, followed by a roll-off. This peak corresponds to the high Q-factor of the crystal. The piezo acts as an differentiator (cap in parallel) and this must be compensated by an integrator. For proper bias, we must add an extra resistor which will create an extra zero. But assuming sufficient gain, the current flows through C gain shielding from cable or n capacitance.

Question 25: Elaborate on acoustic transmission. What is acoustic impedance? what is reflection / transmission.

Answer:

- **Acoustic Impedance (Z_A):** A material property defined as the product of density ρ and sound velocity c_L ($Z_A = \rho c_L$). It determines how sound propagates. Across/through variable for any system.
- **Reflection/Transmission:** When sound hits an interface between two materials with different impedances (Z_1, Z_2), part of the energy is reflected and part is transmitted. The reflection coefficient depends on the mismatch $(Z_2 - Z_1)/(Z_2 + Z_1)$.

Question 26: Explain echography. Give some applications. How can we measure depth and material at the same time?

Answer: Echography transmits an ultrasonic pulse and listens for reflections (echoes) from interfaces.

- **Applications:** Medical imaging (fetus), Non-Destructive Testing (cracks in metal).
- **Depth/Material:** Depth is measured by the Time-of-Flight (ToF) of the echo ($d = c \cdot t/2$). Material properties (impedance change) are inferred from the *amplitude* of the reflected echo.

Question 27: Why does a doctor use a gel in echography. Explain the properties of the gel. How would the gel change for measuring cracks in metal (no numbers / just principle)

Answer: Gel is an impedance matching layer. Without it, the large impedance mismatch between the probe (quartz) and air would cause total reflection, and no sound would enter the body.

- **Properties:** The gel has an acoustic impedance between that of the probe and the skin ($Z_{gel} \approx \sqrt{Z_{probe} Z_{skin}}$) to maximize transmission.
- **For Metal:** A different coupling medium (oil or water) with a much higher impedance is needed to match the high impedance of metal.

Question 28: Explain the use of the doppler effect in the case of a flowmeter.

Answer: A flowmeter transmits sound into a flowing fluid. Particles or bubbles in the fluid reflect the sound. Because the particles are moving, the reflected frequency is shifted (Doppler shift) proportional to the flow velocity. Measuring the frequency shift Δf allows calculating the flow speed.

Question 29: Use the theory of solid-state physics to explain absorption and transmission of light by specific molecules.

Answer: Molecules have discrete energy bands. Absorption occurs only if the energy of an incoming photon (hf) exactly matches the energy gap ($E_2 - E_1$) required to excite an electron to a higher energy state. If the energy does not match, the photon is transmitted. This creates a unique absorption spectrum (fingerprint) for each molecule.

Question 30: Explain the working principle of a LED.

Answer: An LED is a forward-biased p-n junction. Electrons from the n-side and holes from the p-side are injected into the depletion region where they recombine. This recombination releases energy in the form of photons with energy equal to the bandgap.

Question 31: Discuss the working principle of a photon detector.

Answer: A photon detector (like a photodiode) works on the reverse principle of an LED. Incident photons with energy greater than the bandgap excite electrons from the valence band to the conduction band, creating electron-hole pairs. In a reverse-biased junction, these carriers are swept away by the field, creating a measurable photocurrent.

Question 32: What is the difference between a bolometer and a photon detector.

Answer:

- **Bolometer:** A thermal detector. It absorbs radiation, heats up, and measures the resistance change (using a thermistor). It is broadband (detects all wavelengths) but slow. They are painted in black to increase their heat absorption.
- **Photon Detector:** A quantum detector. Photons directly knock electrons into the conduction band. It is very fast and wavelength-selective (depends on bandgap), but usually requires cooling to reduce thermal noise.

Question 33: Draw the basic schematic of a readout of a photon detector.

I think this is the answer but couldn't find anything precise in the course.

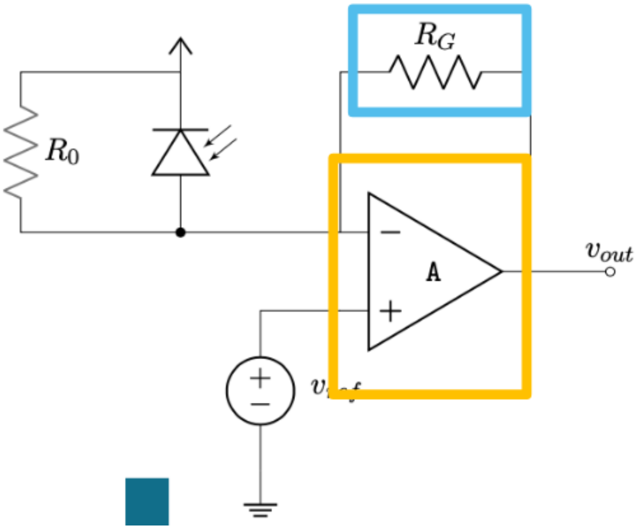
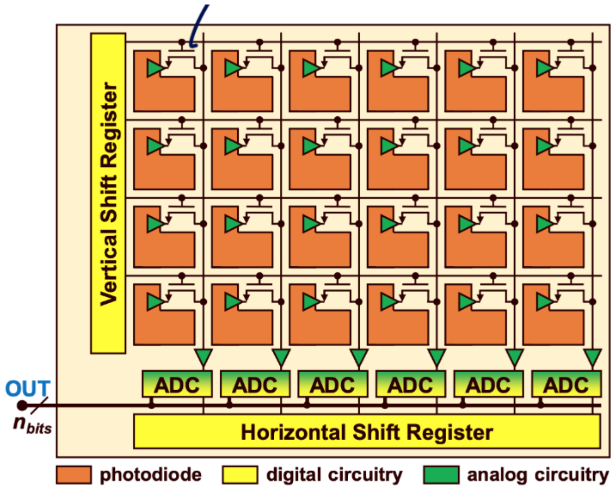
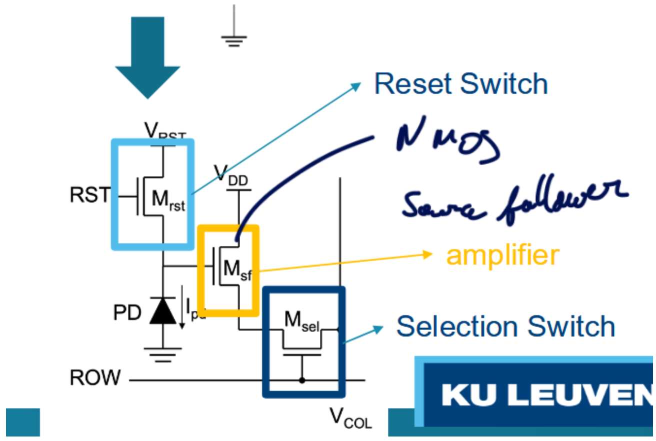


Figure 15: Photodiode readout

Answer: The basic readout is a **Transimpedance Amplifier (TIA)**. The photodiode acts as a current source connected to the inverting input of an op-amp. A feedback resistor R_f connects the output to the input. The output voltage is $V_{out} = -I_{photo} \cdot R_f$.

Question 34: Discuss the imager matrix structure and the readout schematic.





Answer: Imagers consist of a matrix of Active Pixels. Each pixel contains a photodiode and transistors for reset, selection, and amplification (Source Follower). Readout is done row-by-row (Rolling Shutter). When a row is selected, the voltage from each pixel is transferred to column ADCs for digitization.

Question 35: Give 3 examples of optical sensors : one where we measure a difference in source, one where we measure a difference in medium, one where we measure ToF

Answer:

1. **Source:** NDIR gas analyzer (measures absorption of specific wavelengths by gas especially hydrocarbons).
2. **Medium:** Liquid level sensor using a prism (measures change in reflection coefficient/refractive index of the medium surrounding the prism).
3. **ToF:** LiDAR (Light Detection and Ranging) measures distance by time-of-flight of a laser pulse. Or the Interferometer.

Question 36: Give the difference between a Lidar and Radar system.

Answer: LiDAR uses laser light (optical micro meter), while RADAR uses radio waves (cm). LiDAR has much shorter wavelengths, allowing for higher resolution and more precise imaging (beam steering, ...), but is more susceptible to weather (fog/rain). RADAR has longer range and better weather penetration but lower resolution and usually requires larger antenna.

Question 37: Explain the working principle of an interferometer.

Answer: An interferometer (e.g., Michelson) splits a light beam into two paths: a reference path and a measurement path. When reflected back and recombined, the beams interfere constructively or destructively depending on the path difference relative to the wavelength λ. This allows measuring displacement with sub-wavelength precision.

Question 38: Give 3 examples of a time-of-flight sensor.

Answer:

1. Ultrasonic distance sensor (echography).
2. LiDAR.
3. Inferometer

Question 39: Give 2 extensive and concrete examples of digital signal compensation used in a sensor.

Answer:

1. **Hall Sensor Temperature Compensation:** A Hall sensor's sensitivity decreases with temperature. A DSP measures the temperature using an integrated sensor and digitally scales the Hall reading using a polynomial model (e.g., $1 + aT + bT^2$) to flatten the response.
2. **Image Sensor Corrections:** DSP corrects for "Dead Pixels" (interpolating from neighbors), "Dark Current" (subtracting a dark frame offset), and "Fixed Pattern Noise" (correcting pixel-to-pixel gain variations).

Question 40: Why do we want to fit the input range of the ADC ? Elaborate on the speed-power-accuracy trade-off. Bonus question : how can we use an adc to remove the influence of the drive voltage of a Hall sensor.

Answer:

- **Range:** To maximize the Signal-to-Noise Ratio (SNR). If a small signal uses only a fraction of the ADC range, resolution is wasted and quantization noise dominates.
- **Trade-off:** Accuracy costs power. The relationship $\frac{\text{Speed}\cdot(\text{Accuracy})^2}{\text{Power}} = C_T$ implies that increasing accuracy (SNR) requires a quadratic increase in power, while speed scales linearly.
- **Bonus:** Use a **ratiometric measurement**. If the Hall sensor is powered by V_{drive} and the ADC uses V_{drive} as its reference voltage (V_{ref}), the term V_{drive} appears in the numerator (signal) and denominator (ADC scaling), cancelling out any fluctuations in the drive voltage.

Gemini Pro answer

Based on the provided course material `cursus-1.pdf`, here are the in-depth and precise answers to the exam questions from `exam_questions.pdf`.

5.0.1. 1. Theory

Question 1: Discuss resolution in measurement systems, give 2 examples. What is the influence on noise?

- **Resolution:** Resolution is defined as the maximum variation at the input of a measurement system which does not trigger a change at the output. It represents the smallest detectable change in the physical quantity being measured. It is often expressed as a percentage of the input span or in bits.
- **Examples:**

1. **Wire-wound potentiometer:** The resolution is determined by the spacing of the wire windings. The wiper moves from one winding to the next, creating discrete steps in resistance rather than a continuous change.
2. **Analog-to-Digital Converter (ADC):** The resolution is typically defined by the number of bits (). For an input range of , the smallest detectable step is .

- **Influence on Noise:** Resolution is closely linked to quantization noise. A limited resolution introduces an error that can be modeled as noise (quantization noise). For an ideal quantizer, this noise power is . Higher resolution (smaller steps) reduces this quantization noise, thereby improving the signal-to-noise ratio (SNR).

Question 2: Describe hysteresis. Give one example. Explain how feedback can be used to reduce hysteresis.

- **Hysteresis:** Hysteresis is a “memory” effect where the output of a sensor differs for the same input value depending on whether the input was increasing or decreasing. It is defined as the difference .
- **Example:** A **gear system** converting linear to angular movement exhibits hysteresis due to backlash (gaps between gear teeth). When changing direction, the input must cross the gap before the output starts moving, causing a different path for forward and backward motions. Ferromagnetic materials also show hysteresis in their magnetization curves.
- **Feedback:** High-gain feedback reduces errors in the forward path. If a sensor with hysteresis is placed in the forward path of a control loop with a high loop gain, the effect of hysteresis is divided by the loop gain, effectively linearizing the system. However, if the sensor is in the feedback path, its errors (including hysteresis) directly affect the output.

Question 3: Elaborate on non-linearity. Which measurement characteristics describe non-linearity. Explain how feedback can be used to reduce non-linearity.

- **Non-linearity:** This is the deviation of a system’s input-output relationship from an ideal straight line.
- **Characteristics:**

1. **Maximum non-linearity error ():** The maximum deviation between the actual curve and the ideal line, often expressed as a percentage of the span.
2. **Integral Non-Linearity (INL):** In digital systems, this corresponds to the maximum deviation from the ideal transfer line.
3. **Differential Non-Linearity (DNL):** The variation in step size for adjacent digital codes; it measures the deviation of a single step from the ideal 1 LSB width.

4. **Harmonic Distortion (THD):** In the frequency domain, non-linearity creates harmonics () for a sine wave input.

- **Feedback:** Similar to hysteresis, negative feedback reduces non-linearity. If a non-linear component is in the forward path and is surrounded by a high-gain feedback loop, the overall system transfer function approximates the inverse of the feedback network (). Thus, if the feedback network () is linear, the output () becomes a linear function of the input (), suppressing the non-linearity of the forward component ().

Question 4: Discuss the difference between rise time and settling time for a second order measurement system.

- **Rise Time ():** The time it takes for the system's response to rise from a low value (usually 0% or 10%) to a high value (usually 90% or 100%) of the final step value for the first time. It indicates the speed of the system's initial reaction.
- **Settling Time ():** The time required for the response to enter and stay within a specified error band (e.g.,) of the final steady-state value. It accounts for both the initial rise and the decay of any oscillations (ringing).
- **Key Difference:** Rise time measures the *fastest edge*, while settling time measures how long it takes to actually *reach stability*. A system can be fast (short rise time) but take a long time to settle if it is underdamped and oscillates significantly.

Question 5: Discuss the importance of differential systems. Discuss non-linearity and environmental interferers.

- **Importance:** Differential systems measure the difference between two inputs () rather than a single-ended value relative to ground.
- **Non-linearity:** Differential systems inherently cancel out **even-order harmonics** (distortions). If the system is symmetric, the even terms in the Taylor series expansion of the two inputs cancel each other out (), leaving only odd harmonics. This reduces total harmonic distortion.
- **Environmental Interferers:** Differential systems reject **common-mode interference**. Environmental noise (like electromagnetic interference from mains voltage) often affects both signal lines equally (). Since the output is the difference (), this common-mode noise is subtracted out, improving immunity to external errors.

Question 6: Explain aliasing. Couple this to kT/C noise. When having a resistive sensor of MOhm filtered with a capacitor of 100pF, why would we need to take when we sample at 1kHz?

- **Aliasing:** Aliasing occurs when a signal is sampled at a frequency () lower than twice its bandwidth (Nyquist rate). Frequencies above "fold" back into the baseband, becoming indistinguishable from lower frequencies.
- **kT/C Noise:** This is the total integrated thermal noise power in a resistor-capacitor (RC) circuit, given by .
- **Coupling:** In the example, and . The bandwidth (corner frequency) of the RC filter is .
- **Why take kT/C:** We sample at , which gives a Nyquist frequency of 500 Hz. The filter bandwidth (10 kHz) is much higher than the Nyquist frequency. This means noise between 500 Hz and 10 kHz is *not* filtered out before sampling; instead, it aliases (folds) back into the 0–500 Hz baseband. All the noise power from the wide bandwidth accumulates in the baseband. Therefore, the total noise variance seen by the sampler is the full integrated noise , not just the noise within the 500 Hz bandwidth.

Question 7: Long term drift. Explain why we can accelerate the drift measurements. Give 2 different drift tests.

- **Accelerating Drift:** Drift is often caused by chemical or physical aging processes (e.g., oxidation, diffusion) which follow the Arrhenius equation. These processes are temperature-dependent. By increasing the temperature, the reaction rates increase exponentially, allowing us to simulate long-term aging in a short time.

- **Drift Tests:**

1. **High Temperature Operating Life (HTOL):** The device is operated at an elevated temperature (e.g.,) for an extended period (e.g., 1000 hours) to simulate years of operation at room temperature.
2. **Temperature Cycling Test (TCT):** The device is cycled between extreme temperatures (e.g., to). This stresses the mechanical interfaces (like bond wires or packaging) due to thermal expansion mismatches, revealing drift caused by mechanical fatigue or delamination.

Question 8: Explain the difference between white noise and pink noise. Explain why we cannot filter 1/f noise.

- **White Noise:** Noise with a constant power spectral density across all frequencies (flat spectrum). It is caused by thermal motion of electrons (Johnson-Nyquist noise).

- **Pink Noise (1/f Noise):** Noise where the power spectral density is inversely proportional to frequency (). It dominates at low frequencies.

- **Filtering:** We generally cannot filter 1/f noise using standard linear filters (low-pass/high-pass) because it is present in the same low-frequency band as the measurement signal (DC or slow-varying signals). Filtering the noise would also filter out the signal of interest. Techniques like **chopping** or **auto-zeroing** (modulation) are required to move the signal to a higher frequency where white noise dominates, effectively separating it from the 1/f noise.

Question 9: Why can we represent resolution as quantization noise? Can we filter quantization noise? Elaborate.

- **Resolution as Noise:** Quantization maps a continuous input to discrete steps. The error between the actual input and the quantized output () behaves like a random variable uniformly distributed between and . This error adds uncertainty to the measurement, statistically acting like an additive noise source with variance .

- **Filtering:** Yes, quantization noise can be filtered *if* the signal is oversampled. Quantization noise is roughly white (spread over the Nyquist bandwidth). By sampling much faster than the signal bandwidth and then applying a digital low-pass filter (decimation), we remove the noise power outside the signal band, effectively increasing the resolution (ENOB).

Question 10: Give a block diagram of a generalized measurement chain. Discuss gain distribution. Why would we sometimes split up the gain in several amplifiers in series?

- **Block Diagram:** Input (True Value) **Sensing Element Signal Conditioning** (Amplification, Filtering) **ADC/Observation** DSP Output.

- **Gain Distribution:** Gain should be distributed to optimize the **Signal-to-Noise Ratio (SNR)** and prevent **saturation**.

- **Splitting Gain:**

1. **Noise:** High gain in the first stage is desirable to suppress the noise contribution of subsequent stages (Friis formula for noise).

2. **Saturation/Offset:** If the first stage has very high gain, it might amplify the sensor's DC offset causing the amplifier to saturate (hit the supply rails). Splitting gain allows us to filter out the offset (e.g., using a capacitor or DAC) between stages before applying the remaining gain.

3. **Bandwidth:** Amplifiers have a Gain-Bandwidth Product (GBW). Splitting gain allows each stage to operate with lower gain, thereby maintaining a higher total bandwidth.

Question 11: Explain chopping. Why would we use this in measurement systems? What is the origin of chopping ripple?

- **Chopping:** Chopping is a modulation technique. The input DC (or low-frequency) signal is multiplied by a high-frequency square wave (chopper), moving the signal to the chopping frequency. After amplification, it is demodulated back to DC.
- **Why use it:** It is used to eliminate **offset** and **1/f noise** from the amplifier. Since the amplifier's offset/noise remains at DC (or low frequency) while the signal is at the chopping frequency, the offset/noise can be filtered out after demodulation (or simply doesn't get demodulated back to DC in the same way).
- **Chopping Ripple:** The origin is the limited bandwidth or delay of the amplifier. When the square wave is amplified, the edges might be rounded or phase-shifted. When demodulated, these imperfections do not perfectly cancel, leaving residual spikes or "ripple" at the chopping frequency.

5.0.2. 2. Measurement Chain & Errors

Question 12: What is aliasing? How can we avoid this? What is the consequence for a measurement system?

- **Aliasing:** As defined in Q6, it is the folding of high-frequency signals/noise into the baseband during sampling.
- **Avoidance:** It is avoided by enforcing the **Nyquist-Shannon sampling theorem**: sample at a rate . In practice, an **anti-aliasing filter** (low-pass filter) is placed before the ADC to attenuate frequencies above .
- **Consequence:** Aliasing causes **distortion** and **errors** that cannot be removed after sampling. High-frequency noise or interference appears as valid low-frequency data, corrupting the measurement accuracy.

Question 13: Discuss calibration what will limit the accuracy? why would we need to take a long calibration time into account? Give an example why knowledge on the sensor helps the calibration process.

- **Accuracy Limit:** Calibration accuracy is limited by the **reference standard's accuracy** and the **noise** during the calibration measurement. You cannot calibrate to an accuracy better than the noise floor at the time of calibration.
- **Long Time:** To minimize the impact of noise during calibration, we need to average the measurement over a long time. Averaging samples reduces the standard deviation of the noise by . Thus, high accuracy requires long measurement times.
- **Sensor Knowledge:** Knowing the sensor's physics helps select the correct **model** (e.g., polynomial vs. exponential). For a thermistor, knowing it follows an exponential law () allows for accurate calibration with fewer points than blindly fitting a high-order polynomial, which might oscillate or fit noise.

Question 14: Explain the difference between compensation and calibration.

- **Calibration:** The process of determining the relationship between the sensor output and the true input value, often by comparing against a known standard. It involves calculating coefficients (gain, offset) to correct the static transfer function.
- **Compensation:** The active correction of errors caused by *secondary* variables (modifying inputs) like temperature or supply voltage. For example, using a temperature sensor to adjust the reading of a pressure sensor to account for its temperature sensitivity. Compensation corrects for changing environmental conditions, whereas calibration corrects for manufacturing tolerances.

Question 15: Draw the schematic of an instrumentation amplifier. Elaborate on its most important properties.

- **Schematic:** An instrumentation amplifier typically consists of 3 op-amps: two input buffers (non-inverting) creating a differential input stage with gain, followed by a difference amplifier. (Refer to Figure 8.3 in the text).

- **Properties:**

1. **High Input Impedance:** The inputs go directly to the gates/bases of the input op-amps, not loading the sensor.
2. **High CMRR:** It strongly rejects common-mode signals (noise) while amplifying the differential signal.
3. **Gain:** The gain is set by a single external resistor (), making it easy to adjust without affecting the matched resistors that determine CMRR.

Question 16: Explain the non-linearity of a Wheatstone bridge with only 1 sensing element. Discuss how to improve this. Give an example on how we could create a measurement system with 2 resistive sensors with inverted sensitivities.

- **Non-linearity:** A single-element bridge (quarter bridge) has an output . The term in the denominator causes non-linearity.
- **Improvement:** Use a **current drive** instead of voltage drive (), or use **feedback** to keep the bridge balanced.
- **2 Sensors (Inverted):** Use two sensors with opposite sensitivities (and), e.g., strain gauges on opposite sides of a bending beam (one in tension, one in compression). Placing them in the same leg (half-bridge) or opposite legs effectively linearizes the output because the non-linear terms in the numerator and denominator cancel out or the sensitivity doubles while maintaining symmetry.

Question 17: Describe the design of an oscillator as a sensor readout interface for a reactive sensor element.

- **Design:** A reactive sensor (capacitor or inductor) is placed in the feedback network of an active circuit (like an op-amp or comparator) to form an oscillator.
- **Principle:** The oscillation frequency is determined by the resonance of the LC or RC network (). As the sensor value (or) changes, the frequency changes.
- **Readout:** The output is a frequency, which can be digitized directly by a counter/timer. This eliminates the need for a conventional ADC, providing a direct "sensor-to-digital" interface.

5.0.3. 3. Sensors

Question 18: Explain the working principle of a strain gauge.

- **Principle:** A strain gauge relies on the **piezoresistive effect** and geometry change. When a conductive material is stretched (strained), its length () increases and cross-sectional area () decreases (Poisson effect), increasing resistance ().
- **Gauge Factor:** The sensitivity is described by the Gauge Factor (), which combines geometric changes and the change in resistivity () due to stress.

Question 19: Explain the working principle of an Hall sensor. Bonus: explain spinning and why we can spin a Hall sensor and not another Wheatstone bridge.

- **Principle:** The Hall effect occurs when a current flows through a conductor in a magnetic field. The Lorentz force deflects charge carriers to one side, creating a voltage () perpendicular to both current and field.
- **Spinning:** A technique to remove offset. The current terminals and voltage measurement terminals of the Hall plate are cyclically swapped (rotated 90 degrees). The magnetic signal polarity remains the same, but the resistive offset polarity flips. Averaging the phases cancels the offset.

- **Why Hall:** A Hall plate is a symmetric 4-terminal device where input/output ports are geometrically identical and interchangeable. A standard Wheatstone bridge has distinct components for each leg; "spinning" it would physically rewire the resistors, which doesn't inherently cancel mismatch offsets in the same way (though chopping is similar).

Question 20: Why do we need at least 2 temperature sensors for getting a high accuracy in a MoX sensor?

- **Reason:** Metal Oxide (MoX) gas sensors use a heater to operate at high temperatures.

1. One temperature sensor is needed to control the **heater temperature** precisely, as gas sensitivity is highly temperature-dependent.
2. A second sensor measures the **ambient temperature** to compensate for environmental drifts and the temperature coefficient of the gas sensing layer itself. (Note: This is an inference based on general sensor fusion principles in section 2.4.2 and temperature compensation in 7.5. The text explicitly mentions temperature compensation for Hall sensors and general thermal dependencies).

Question 21: Explain a capacitive pressure sensor. Why do we need a CDAC?

- **Principle:** It consists of a diaphragm that acts as one plate of a capacitor. Pressure deforms the diaphragm, changing the distance () between plates, thus changing capacitance ().
- **CDAC (Capacitive DAC):** The sensor has a large "rest" capacitance () and a small variation (). A CDAC is used in the readout circuit to subtract the large (offset removal). This allows the amplifier to focus only on the small , preventing saturation and maximizing dynamic range.

Question 22: Explain how an accelerometer works. What is the difference between a capacitive accelerometer and a piezoresistive based accelerometer?

- **Principle:** An accelerometer is a mass-spring-damper system. Acceleration acts on the proof mass (), causing a displacement () against the spring (). Measuring yields the acceleration.
- **Capacitive:** Measures displacement by the change in capacitance between the moving mass and fixed electrodes. It is typically more sensitive and stable for low-g static acceleration (like tilt).
- **Piezoresistive:** Measures the stress in the spring (support beams) using piezoresistors. It is often simpler but can be temperature sensitive.

Question 23: Give 3 examples of capacitive sensors: one where we measure a difference in distance, one in dielectric, one in area.

- **Distance (): Pressure sensor** (diaphragm moves) or **Microphone**.
- **Dielectric (): Humidity sensor**. The polymer dielectric absorbs water from the air, changing its permittivity.
- **Area (): Level sensor** (liquid rising between plates changes effective area) or **Lateral accelerometer** (comb fingers sliding past each other).

5.0.4. 4. Advanced Sensors & Acoustics

Question 24: Draw a typical piezo crystal readout. Explain the bode plot of a typical piezo crystal readout.

- **Readout:** A charge amplifier is typically used. The piezo is a capacitor () generating charge. The readout uses an op-amp with a feedback capacitor () to convert charge to voltage ().

- **Bode Plot:** It behaves like a band-pass filter. At low frequencies, the leakage resistance dominates (high-pass behavior). At resonance, there is a sharp peak (determined by the mechanical Q-factor). Above resonance, the response flattens or rolls off depending on the damping.

Question 25: Elaborate on acoustic transmission. What is acoustic impedance? what is reflection / transmission.

- **Impedance ():** Acoustic impedance is the resistance of a medium to sound flow, defined as (density speed of sound).
- **Reflection/Transmission:** When sound hits an interface between two media (), part is reflected and part transmitted.
- **Reflection Coefficient ():** . Large impedance mismatch causes high reflection.
- **Transmission Coefficient ():** . To maximize transmission (e.g., from sensor to body), a matching layer is used.

Question 26: Explain echography. Give some applications. How can we measure depth and material at the same time?

- **Echography:** Uses the pulse-echo principle. A piezo transducer emits a high-frequency sound pulse. The pulse reflects off internal interfaces (impedance changes) and returns. The time of flight determines depth ().
- **Applications:** Medical imaging (fetus, organs), non-destructive testing (detecting cracks in metal).
- **Depth & Material:**
- **Depth:** From Time of Flight (ToF).
- **Material:** From the **amplitude** of the reflection. A stronger reflection indicates a larger difference in acoustic impedance, characterizing the material interface.

Question 27: Why does a doctor use a gel in echography? Explain the properties of the gel. How would the gel change for measuring cracks in metal?

- **Why Gel:** Air has a very low impedance compared to body tissue (). Without gel, almost 100% of the ultrasound would reflect at the skin surface (impedance mismatch). Gel acts as a **coupling medium** or matching layer.
- **Properties:** The gel's acoustic impedance is designed to match that of human skin/tissue to maximize transmission.
- **For Metal:** The coupling medium (gel/water) would need an impedance closer to that of the metal (or the transducer wedge) to facilitate energy transfer, although perfect matching to metal with a liquid is difficult. Water or oil is often used.

Question 28: Explain the use of the doppler effect in the case of a flowmeter.

- **Principle:** Ultrasonic waves are transmitted into a flowing fluid. Particles in the fluid reflect the sound. The frequency of the reflected sound is shifted due to the particle velocity (): . Measuring the frequency shift () gives the flow velocity.

5.0.5. 5. Optics & Systems

Question 29: Use the theory of solid-state physics to explain absorption and transmission of light by specific molecules.

- **Theory:** Electrons in molecules occupy discrete energy levels (orbitals). Light consists of photons with energy .

- **Absorption:** A molecule absorbs a photon only if the photon's energy exactly matches the energy gap between two electron states (). This excites the electron to a higher energy level. This creates a unique "fingerprint" absorption spectrum for each molecule.

- **Transmission:** Wavelengths that do not match an energy gap pass through (are transmitted). .

Question 30: Explain the working principle of a LED.

- **Principle:** A Light Emitting Diode is a p-n junction. When forward biased, electrons from the n-side recombine with holes from the p-side in the depletion region. This recombination releases energy in the form of photons (spontaneous emission). The wavelength depends on the bandgap energy of the semiconductor material.

Question 31: Discuss the working principle of a photon detector.

- **Principle:** A photon detector (e.g., photodiode) works on the photoelectric effect. A photon with energy greater than the semiconductor bandgap strikes the material, generating an electron-hole pair. In a p-n junction (often reverse-biased), the electric field sweeps these carriers away, creating a measurable current (photocurrent) proportional to light intensity.

Question 32: What is the difference between a bolometer and a photon detector.

- **Photon Detector:** Quantum detector. Reacts to individual incident photons generating charge carriers. Fast, wavelength-dependent (bandgap).
- **Bolometer:** Thermal detector. Absorbs radiation, which heats up the element. The temperature rise causes a change in resistance. It is generally slower (thermal mass) but has a broad, flat spectral response (wavelength independent).

Question 33: Draw the basic schematic of a readout of a photon detector.

- **Schematic:** A **Transimpedance Amplifier (TIA)** is typically used. The photodiode is connected to the inverting input of an op-amp. A feedback resistor () connects output to input.
- **Operation:** The photodiode current () flows through , creating an output voltage . This converts current to voltage while keeping the diode bias constant (virtual ground).

Question 34: Discuss the imager matrix structure and the readout schematic.

- **Matrix:** Pixels are arranged in rows and columns.
- **Readout:**
- **Passive Pixel Sensor (PPS):** One transistor per pixel acting as a switch. Slow, high noise.
- **Active Pixel Sensor (APS/CMOS):** Each pixel contains a photodiode and its own amplifier (3T or 4T structure). This allows charge-to-voltage conversion *inside* the pixel, reducing noise and allowing faster row-by-row readout.

Question 35: Give 3 examples of optical sensors: one where we measure a difference in source, one where we measure a difference in medium, one where we measure ToF.

- **Source: Pyrometer** (measures intensity/spectrum of the blackbody radiation source to determine temperature).

- **Medium: Gas absorption sensor** (measures light attenuation through a gas to determine concentration).

- **ToF: LiDAR** (measures time for a light pulse to reflect back).

Question 36: Give the difference between a Lidar and Radar system.

- **LiDAR:** Uses **Light** (Laser). Shorter wavelength (nm to m). Higher resolution, better for 3D mapping small details, but affected by weather (fog/rain).
- **Radar:** Uses **Radio** waves (cm to mm). Longer wavelength. Lower resolution, but works well in poor weather/visibility conditions.

Question 37: Explain the working principle of an interferometer.

- **Principle:** (e.g., Michelson Interferometer). A light beam is split into two paths (reference and measurement). They reflect back and recombine.
- **Effect:** The difference in path length causes a phase shift. When recombined, constructive or destructive interference occurs. The intensity changes cyclically with distance changes of , allowing for extremely precise distance measurements.

Question 38: Give 3 examples of a time-of-flight sensor.

- 1. **LiDAR** (Optical).
- 2. **Ultrasonic Range Finder** (Acoustic).
- 3. **Radar** (RF).

Question 39: Give 2 extensive and concrete examples of digital signal compensation used in a sensor.

- 1. **Temperature Compensation of a Hall Sensor:** A Hall sensor's sensitivity decreases with temperature. A DSP uses a separate temperature sensor reading. It applies a polynomial correction formula () to the Hall output to cancel the drift.
- 2. **Non-linearity Linearization:** A sensor with a known non-linear transfer function (e.g., thermistor log response) is connected to a DSP. The DSP applies the inverse function (mathematically or via a Look-Up Table) to the digital data, outputting a linear temperature reading.

Question 40: Why do we want to fit the input range of the ADC? Elaborate on the speed-power-accuracy trade-off. Bonus: how can we use an ADC to remove the influence of the drive voltage of a Hall sensor.

- **Input Range:** To maximize **Signal-to-Noise Ratio (SNR)**. If the signal is too small, resolution is wasted. If too large, it clips (saturates). Matching the signal to the ADC full-scale range utilizes all available bits.

- **Trade-off:** (). To increase accuracy (SNR) by 1 bit (6dB, factor of 2 in voltage), we need 4x the signal power (or 4x averaging time, reducing speed). Higher speed requires more power to charge capacitors quickly.

- **Bonus (Hall/ADC):** Use a **ratiometric measurement**. Power the Hall sensor and the ADC reference voltage () from the *same* supply source. The Hall voltage is proportional to the supply (). The ADC output is proportional to . The term cancels out, removing the error caused by supply voltage drifts.

6. License

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